Unit 5 – Polynomial Functions *Accelerated II/III*

Classifying Polynomials & Function Operations (Arithmetic)

Polynomial Function: $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x^1 + a_0$

**Exponents are all _____ numbers and coefficients are all _____ numbers.

**Standard Form: terms are written in descending order of _____ from left to right.

Example 1: A) $f(x) = 4x + 2$

degree = _____

leading coefficient = _____

constant = _____

B) $f(x) = 6x^4 + 2x^3 - 4x^2 + 5x - 10$

degree = _____

leading coefficient = _____

constant = _____

Degree	Type	Example	# of Terms	Name Using # of Terms
0		$f(x) = 5$		monomial
1		$f(x) = -2x$		
		$f(x) = \frac{1}{4}x^2 + 3x + 1$		trinomial
3		$f(x) = -2x^3 + 2x - 9$		
		$f(x) = 4x^4 - 6x^2$		binomial
5		$f(x) = 5x^5 - 4x^4 + 3x^3 - 2x^2 + 1$		polynomial of 5 terms

Example 2: Is it a polynomial function?

A) $f(x) = x^2 + 3x^4 + 1$

B) $f(x) = \pi x^2 + .45x^5 - \sqrt{5}x$

C) $f(x) = x^{-1} + x + 10$

D) $f(x) = 12x + x^4 + 5^x$

Example 3: Write each polynomial in standard form. Then classify it by degree and by number of terms.

A) $9 + x^3$

B) $x^3 - 2x^2 - 3x^4$

C) $4x - 6x + 5$

D) $3x^3 + x^2 - 4x + 2x^3$

Function Operations: Let *f* and *g* be any two functions. A new function *h* can be defined by:

Addition: $(f+g)(x) = f(x)+g(x)$

Subtraction: $(f-g)(x) = f(x)-g(x)$

Multiplication: $(f \bullet g)(x) = f(x)g(x)$

Division: $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)};\ g(x) \neq 0$

Composition: (we will discuss later)

Example 4: Complete the following operations given $f(x) = 2x^3 - 3x^2 - 18x - 8;\ g(x) = x - 4$

A) $(f+g)(x) =$

B) $(f-g)(x) =$

C) $(f \bullet g)(x) =$

D) $\left(\frac{f}{g}\right)(x) =$

Example 5: Find the following:

A) $f(x) = 3x^3 - 18x^2 + 24x;\ f\left(\frac{x}{2}\right)$

B) $f(x) = -2x^2 + 8x - 8;\ f(a-3)$